Ship Classification in SAR Images With Geometric Transfer Metric Learning

Yongjie Xu, *Student Member, IEEE* and Haitao Lang, *Member, IEEE*

***Abstract*— There are still many challenges to be resolved in the task of ship classification in synthetic aperture radar (SAR) images, such as limited number of labeled samples in SAR domain, large variance in the same subcategory, small variance among different subcategories, etc. Transfer metric learning (TML) has the potential to mitigate those issues in the domain of interest (target domain, TD) by leveraging knowledge/information from other related domains (source domain, SD). In this article, we proposed a novel TML method, termed as geometric transfer metric learning (GTML), which achieves discriminative information preservation (DIP), geometric structure preservation (GSP), and handles the domain shift (DS) simultaneously by integrating pairwise constraints (PC), joint distribution adaptation (JDA), and manifold regularization (MR) into a unified optimization function, aiming to make full use of their complementarity to improve SAR ship classification performance. In practice, we proposed two simple but effective optimization strategies, termed as GTML-A and GTML-R, to construct optimization function. We also proposed two solutions for two typical real-world application scenarios, that is, the task of: 1) zero-labeled sample (ZLS) and 2) scarce-labeled samples (SLS) in SAR domain. The experiments conducted on both tasks show that the proposed GTML outperforms most of state-of-the-art methods. Code is available at https://github.com/sky-Yongjie-Xu/geometric-transfer-metric-learning.**

***Index Terms*— Geometric structure preservation (GSP), ship classification, synthetic aperture radar (SAR), transfer metric learning (TML).**

I. Introduction

AS THE most important carrier of human activities at sea, ships play an important role in a variety of maritime matters including border control, environmental protection, traffic surveillance, search and rescue, and so on. With respect to these needs, it is very important to monitor maritime ships by any possible means [1]. The role of the space-borne synthetic aperture radar (SAR)[1] in maritime ship monitoring has become widely accepted, since space-borne SAR is able to provide a powerful local/global monitoring capability to allow observation over wide swaths, independent of the active cooperation of ships, and is less restricted by meteorological phenomena and atmospheric conditions [2]. With the launch of the new generation of satellite missions, the number of moderate- and high-resolution SAR (MR-SAR and HR-SAR) ship images has gradually increased, making the maritime ship monitoring mission further developed from provide geographic location of the ship target (i.e., ship detection) to identify the category of ship (i.e., ship classification).



Fig. 1. Image classification versus fine-grained image classification. (a) Top line represent the common image classification task, which aims to recognize object from basic-level category, for example, ship, plane, bicycle, car, or motorcycle. (b) Middle line and (c) bottom line represent the fine-grained image classification task in optical images and in SAR images, respectively. Different from the common image classification task, fine-grained image classification aims to recognize different subordinate-level categories, for example, cargo, container, or tanker, belonging to the same basic-level category (ship).

Recent years, the supervised learning methods, including traditional machine learning methods [4]–[8] and newly emerging deep learning methods [9], [10], have show promising results on the task of ship classification in SAR images. However, ship classification in SAR images is by no means a simple task. It remains understudied and still highly challenging. First of all, it is a typical fine-grained classification problem [11]. As shown in Fig. 1, beyond identifying general category, for example, identifying which is a ship, a plane, a bicycle, a car, or a motorcycle from different vehicles [Fig. 1(a)], fine-grained image classification aims to recognize different subordinate-level categories, for example, cargo, container, or tanker, belonging to the same basic-level category (ship) [see Fig. 1(b) and (c)]. SAR image-based fine-grained ship classification [Fig. 1(c)] is more challenging, because: 1) there is only subtle variation in visual appearance among ship subordinate-level categories, which is difficult to recognize even for human experts with domain knowledge, and 2) SAR images can only provide limited discriminative

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The authors are with the College of Mathematics and Physics, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: langht@mail.buct.edu.cn; haitaolang@hotmail.com).

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[1]The names of existing approaches, some proper nouns and algorithms mentioned in this article, and their corresponding abbreviations are listed in Appendix Section.

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information due to the limitation of SAR imaging mechanism, compared to the optical images [Fig. 1(b)]. To cope with the weak discriminative information, most researchers were committed to designing more discriminative features. Xing *et al.* [4] proposed to employ a representative feature vector to construct the dictionary instead of utilizing the image pixels directly, to describe the ship more accurately and to reduce the dimension of the dictionary in sparse representation classification. Lang and Wu [8] proposed naive geometric features (NGFs) with different physical properties and discriminability for ship classification in moderate-resolution SAR images. Lang *et al.* [7] also proposed a joint feature and classifier selection method by integrating the classifier selection strategy into a wrapper feature selection framework. Unfortunately, these metric-based methods predefined the distance metric (e.g., Euclidean distance metric) for artificial features, without mining the potential relationship among feature representations, thus only achieved a limited performance improvement. As far as we know, learning a good distance metric reflecting the underlying structures and distributions of a specific database to improve the discriminative ability of feature representations, may be the key to further improving the performance of these metric-based methods [12]. Xu *et al.* [13] evaluated multiple distance metric learning (DML) methods on SAR ship classification task, and verified the feasibility of DML for SAR image-based fine-grained ship classification, and then they proposed the distribution shift metric learning (DML-ds) to improve the discriminative ability of distance metrics by introducing interclass distribution shift regularization [14], [15].

Another dilemma for ship classification in SAR images based on supervised learning methods is that those methods rely on the availability of sufficient labeled samples to train a classifier. Unfortunately, labeling ships in SAR images is often a time-consuming and costly process that requires laborious expert interpretation and/or expensive ground campaigns, and is even not possible in many real-world application scenarios [16], [17]. To remedy the lack of labeled samples, semisupervised learning methods are proposed to comprehensively utilize the labeled and unlabeled data to generate appropriate classification function [18]. Nevertheless, semisupervised learning assume the distribution of the labeled and unlabeled data is same, which requires all data come from the same acquisition condition (e.g., frequency, polarization, viewpoint, and sensor resolution) which still limits the available amount of data. Compared with labeling ships in SAR images, labeling ships in optical images [see Fig. 1(b)] or collecting labeled ship information provided by the automatic identification system (AIS) from public website is considerably easier. For this reason, the method of transfer learning (TL) and/or domain adaptation (DA) is introduced to improve the ship classification performance in SAR images (target domain, TD) by transferring knowledge from auxiliary domain (source domain, SD) [16], [17], [20]. The ground-breaking research proposed by Lang *et al.* [16] proved that the performance of ship classification in SAR domain can be effectively improved by passing information from AIS domain, with only a few labeled samples in SAR domain. Furthermore, Xu and Lang *et al.* [17] proposed the discriminative adaptation regularization framework-based transfer learning (D-ARTL) to address the problem in case that there is no labeled sample available in SAR domain.

A. Motivation

As aforementioned, DML can effectively solve the problem of weakly discriminative information which is often faced in the task of fine-grained ship classification in SAR images [13]–[15], and TL/DA can mitigate the labeled data scarcity issue in SAR domain with the help of a large amount of labeled samples in the AIS domain [16], [17]. Intuitively, when applying TL/DA to improve DML, it will be able to deal with both weakly discriminative information and label deficiency problems simultaneously. This leads to the so-called transfer metric learning (TML), which has been found to be useful in many applications [21], but has not been used to tackle the challenges of fine-grained ship classification in SAR images as far as we know. Inspired by this, this article is devoted to propose a novel TML method, which can further improve the performance of ship classification in the SAR domain when there are only a few (scarce) samples, or even no labeled samples at all.

Focusing on the task of ship classification in SAR images, when delving into the distribution of transferable features[2] as shown in Fig. 2, we can find severe domain shift (DS) (the marginal distribution and conditional distribution between two domains are significantly different) and very poor feature discrimination (the interclass mixing and intraclass variance are equally obvious). To cope with these problems, the existing DML, TL/DA, and TML methods (we conducted a survey in Section II) often introduce discriminative information preservation (DIP) and/or joint distribution adaptation (JDA) as the optimization component(s) into an objective function to be optimized. Generally speaking, DML methods can automatically learn the distance metrics and implement the DIP by constructing pairwise constraints (PC) or triple constraints (TC) under a supervised learning framework but does not have the ability to handle DS [12], [13]. In contrast, most of TL/DA methods can handle the DS well by conducting JDA, and some of them also consider DIP to further improve performance [19], [20], [22]. Although the newly emerging TML methods aim to employ TL to improve DML, in fact, many methods still fail to solve the problem of DIP and DS at the same time [21]. As a typical example, the robust transfer metric learning (RTML) [23] can effectively mitigate the DS by exploiting knowledge transfer in sample space and feature space, but neglected the importance of DIP. We also found that existing methods including DML, TL/DA, and TML rarely consider the data geometric structure preservation (GSP). The data geometric structure, meaning close samples tend to have the same label, should be preserved when the common latent factors are discovered as the bridge for knowledge transfer. Otherwise, transfer learners based on the discovered bridge

[2]Following [16], [17], in this article, we utilize the AIS domain to assist the ship classification in the SAR domain, and still use NGFs as the transferable feature connecting the two domains.

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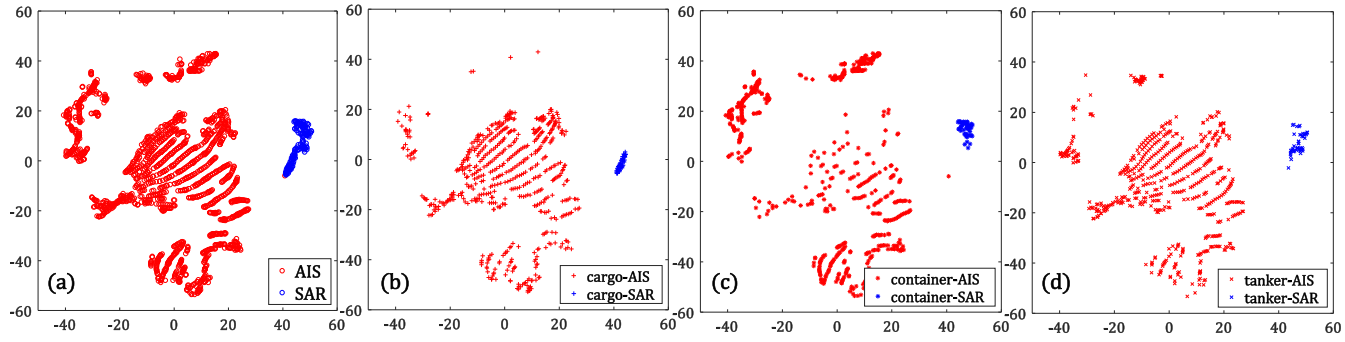


Fig. 2. DS between AIS domain and HR-SAR domain. (a) Original marginal distribution and (b)–(d) corresponding conditional distribution of transferable feature on AIS domain (red) and HR-SAR domain (blue). This visualization is plotted based on t-distributed stochastic neighbor embedding (t-SNE) method [3].

may fail to predict labels smoothly with respect to the geometric structure [24]. Unfortunately, so far, there are few methods to consider DIP, JDA, and GSP simultaneously. For instance, the domain adaptation metric learning (DAML) [25] and metric transfer learning framework (MTLF) [26] can preserve the discriminative information (i.e., DIP) by leveraging PC, but ignored the importance of GSP. The manifold-determinant regularized distance metric learning (M-DML) [27] and manifold regularized transfer distance metric learning (MTDML) [28] explore the local geometric structure of data and make the distance metrics being smooth over the data adjacency graph by introducing manifold regularization (MR), but ignored the importance of JDA.

For the purpose of improving the performance of ship classification in SAR images, we propose a novel method of TML, termed as geometric transfer metric learning (GTML), which achieves DIP, GSP, and handle the DS simultaneously by jointly optimizing PC, JDA, and MR, aiming to make full use of their complementarity. Extensive experiments demonstrate that the proposed GTML outperforms most state-of-the-art methods in the task of SAR ship classification.

B. Contributions

The major contributions of this article are twofold. 1) For the purpose of improving ship classification performance in SAR images, to fully preserve and utilize the discriminative information underlying the data and effectively mitigate the DS between the SAR and AIS domains, this study propose GTML which can achieve DIP, GSP, and handle the DS simultaneously by jointly optimizing PC, JDA, and MR in an objective function. Specially, GTML achieves DIP based on PC, exploits JDA to deal with DS issue between SD and TD, and leverages MR to achieve GSP. In addition, we propose two simple but effective optimization strategies termed as GTML-A and GTML-R, respectively. Compared with the existing methods that can also deal with the problems of DIP, GSP, and DS, under the same performance, the proposed optimization strategy is simpler and more practical. 2) Facing two extremely challenging application scenarios, that is, zero-labeled sample (ZLS) task which there is no labeled sample available in TD, and scarce-labeled samples (SLS) task which there are only a few (in this study, three) labeled samples in TD, we have proposed specific solutions for two tasks. Extensive experiments demonstrate that the performance of ship classification in SAR images can be further improved by the proposed GTML on both tasks and outperforms most state-of-the-art methods.

The remainder of this article is organized as follows: We present a comprehensive survey on the methods of DML, TL/DA, and TML which are related to the work of this article in Section II. A detailed description to the proposed method is presented in Section III, including the mathematical notations and definitions (Section III-A), three optimization components (Section III-B), two optimization strategies (Section III-C), and specific solution for ZLS and SLS task, respectively (Section III-D). In Section IV, we first introduce the comparison methods (Section IV-A), data preparation, and experiment protocol (Section IV-B), then we report and analyze the experimental results in the task of ZLS (Section IV-C) and SLS (Section IV-D), respectively. To present an in-depth analysis, in addition to the result comparison, we also conduct ablation study, feature visualization, confusion matrix analysis, parameter sensitivity analysis, and runtime comparison. Finally, we conclude this study and outlook the future work in Section V.

II. Related Work

A. DML

DML methods, based on the supervised learning framework, can automatically learn the distance metrics describing the underlying relationships between data points to improve the discriminative ability of feature representations [12], [13]. Most DML methods construct PC or TC based on different loss functions, and cooperate with some regularization to learn distance metrics.

Max-min constraints (MMC) [29], discriminative component analysis (DCA) [30], trace discriminative component analysis (TrDCA) [31], Laplacian regularized metric learning (LRML) [32], and DML eigenvalue optimization framework (DML-eig) [33] methods performed PC based on max-min loss (MML). In addition, LRML introduced MR to explore the geometric structure property of nearest points, DML-eig optimized objective function based on eigenvalue optimization technology.

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Regularized distance metric learning (RDML) [34], sparse metring learning (SML) [35], similarity metric learning (Sub-SML) [36], and doublet support vector machine (D-SVM) [37] methods performed PC based on hinge loss (HL). Large margin nearest neighbor (LMNN) [38], triplet support vector machine (T-SVM) [37], and linear ordinal distance metric learning (LODML) [39] methods performed TC based on HL. In addition, RDML is robust to the high-dimensional data, SML introduced $L_{2,1}$-norm to remove outliers, Sub-SML introduced similarity function to improve the robust and discriminative ability of similarity metrics, and LODML introduced local TC (LTC) containing the ordering information to preserve the ordinal relationship and geometric structure.

BoostMetric performed PC based on exponential loss (EL) and learned distance metrics based on boosting techniques [40]. DML using DC programming (DML-DC) performed PC based on ramp loss (RL) and introduced large-margin principle (LMP) to improve the discriminative ability of distance metrics [41]. Information-theoretic metric learning (ITML) minimized the differential relative entropy of the multivariate Gaussians of transformation matrix and identity matrix based on log-determinant divergence (LD), and adopted MML to construct PC constrained distance metrics [42]. DML through maximization of the Jeffrey divergence (DMLMJ) maximized the Jeffrey divergence (JD) of two multivariate Gaussian distributions derived from local PC [43]. Neighborhood component analysis (NCA) minimized the expected leave-one-out error (LOE) of a stochastic nearest neighbor (NN) classifier in the projection space [44].

B. TL and DA

TL and DA can use SD prior knowledge to assist in TD classification task [20], [22]. DA is often regarded as a special case of TL, namely both the SD and the TD share the same classification task [19]. This study focuses on the DA issue, more specifically, focuses on *homogeneous* feature-based DA, which assumes that SD and TD have the same feature space but have different feature distribution. This section mainly reviews the DA methods which are directly related to our study.

1) Unsupervised Domain Adaption: Unsupervised domain adaption (UDA) assumes that SD has a large amount of labeled data, and TD is completely unlabeled data. i) *Distribution adaptation.* Stratified transfer learning (STL) performed conditional distribution alignment (CDA) [45]. Transfer component analysis (TCA) [46], adapting component analysis (ACA) [47], transfer sparse coding (TSC) [48], transfer joint matching (TJM) [49], transfer kernel learning (TKL) [50], latent graph alignment (LGA) [51], and transfer latent representation (TLR) [52] performed marginal distribution alignment (MDA). Among them, ACA introduced Hilbert-Schmidt independence criterion (HSIC) to increase the dependence between SD and TD, TJM introduced $L_{2,1}$ norm to achieve SD instance reweighting (IR), TSC introduced graph regularized sparse coding (GRSC) to achieve GSP and improve the robust of domain-invariant features, LGA minimized the spectrum distance of the graph Laplacians between SD and TD feature embeddings, and TLR introduced linear auto-encoder to reconstruct SD and TD features. JDA [24], visual domain adaptation (VDA) [53], balanced distribution adaptation (BDA) [54], joint geometrical statistical alignment (JGSA) [55], discriminative label consistent domain adaptation (DLC-DA) [56], and weight JDA (W-JDA) [57] performed JDA. In addition, VDA introduced within-class scatter matrix (WSM) to achieve source DIP (SDIP), BDA considered the importance between MDA and CDA, JGSA leveraged WSM and interclass scatter matrix (ISM) to achieve SDIP, introduced target variance maximization (TVM) to achieve GSP, and leveraged subspace divergence minimization (SDM) to further reduce the discrepancy between SD and TD, DLC-DA introduced interclass repulsive force (IRF), and label consistent regression (LCR) to achieve DIP, W-JDA reweighted SD instances by their relevance to the TD instances based on distance measure. ii) *Subspace learning.* Geodesic flow kernel (GFK) integrated an infinite number of subspaces to characterize changes in geometric and statistical properties from SD to TD [58]. Subspace alignment (SA) aligned the base vectors directly [59], and subspace distribution alignment (SDA) further considered distribution alignment between SD and TD [60]. Correlation alignment (CORAL) aligned subspaces based on second-order statistics [61]. Linear-discriminant-analysis-inspired domain adaptation (LDADA) utilized the class mean to learn the class-specific linear projection [62]. Latent sparse domain transfer (LSDT) combined SD and TD data to reconstructs TD data [63]. Low-rank sparse representation (LRSR) leveraged SD data to reconstruct TD data, preserved the local and global structure of domain-invariant subspace based on low-rank constraints (LC) and sparse constraints (Sc), respectively, and adopted LCR to achieve SDIP [64], low-rank discriminative reconstruction (LRDR) [65] further introduced linear entropy (LE) and max-margin classifier (MC) to evaluate the discriminative ability of SD and TD, respectively. iii) *Domain-invariant classifier learning.* Adaptation regularization framework-based transfer learning (ARTL) jointly optimized structural risk minimization (SRM), JDA, and MR to learn a domain-invariant classifier [66], manifold embedded distribution alignment (MEDA) further learned a domain-invariant classifier in Grassmann manifold and performed dynamic JDA to quantitatively account for the relative importance of MDA and CDA [67].

2) Supervised Domain Adaptation (SDA): SDA assumes that SD has a large amount of labeled data and TD has a small amount of labeled data. Domain adaptation manifold alignment (DAMA) leveraged PC and MR to achieve DIP and GSP, respectively, [68]. Max-margin domain transforms (MMDT) jointly optimized transformation matrix that project features from TD to SD and classifier parameters based on misclassification loss [69]. Heterogeneous feature augmentation (HFA) used HL to achieve DIP and combined multikernel learning (MKL) to learn domain-invariant subspaces [70]. Cross-domain landmarks selection (CDLS) identified the contribution of each landmark when matching cross-domain class-conditional data distribution in the derived

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feature space and adopted L_2 norm (L_2) constraint mapping matrix [71].

C. TML

TML combines DML and TL, and uses SD distance metrics or SD data as priori knowledge to help TD learn distance metrics [21].

1) Unsupervised Transfer Metric Learning: Unsupervised transfer metric learning (UTML) assumes TD is completely unlabeled data. DAML [25] and cross-domain metric learning (CDML) [72] performed MDA, and RTML [23] performed JDA. In addition, DAML performed PC based on MML, CDML adopted MR to achieve DIP and GSP, RTML introduced marginalized denoising fashion to reconstruct SD and TD data, and leveraged LC to improve the robust of distance metrics. Information-theoretical learning (ITL) [73] combined mutual information (MI) and clustering method to jointly learn domain invariant features and classifiers.

2) Supervised Transfer Metric Learning: Supervised transfer metric learning (STML) assumes TD has a small amount of labeled data. Log-determinant regularized distance metric learning (L-DML) and M-DML performed PC based on MML, and introduced LD and MR to implement metric approximations of SD and TD, respectively, [27]. Decomposition-based transfer distance metric learning (DTDML) assumed TD distance metrics can be represented as a combination of a series of SD distance metrics, and minimized the Frobenius norm (FN) of SD and TD distance metrics based on RDML framework to achieve knowledge transfer [74]. MTLF performed PC based on MML, and reweighted SD data based on importance sample.

III. Methodology

A. Notations and Definitions

Suppose a domain consists of a d-dimensional feature space and a marginal distribution, and its corresponding task consists of a label space and a conditional distribution. Given a labeled SD $D_s = \{(\boldsymbol{x}_1^s, \boldsymbol{y}_1^s), \ldots, (\boldsymbol{x}_{n_s}^s, \boldsymbol{y}_{n_s}^s)\}$ and an unlabeled TD $D_t = \{(\boldsymbol{x}_1^t), \ldots, (\boldsymbol{x}_{n_t}^t)\}$ for ZLS task or a TD only have a few labeled samples $D_t = \{(\boldsymbol{x}_1^t, \boldsymbol{y}_1^t), \ldots, (\boldsymbol{x}_{n_t}^t, \boldsymbol{y}_{n_t}^t)\}$ for SLS task. This article focuses on the *homogeneous* TML problem, thus assumes that the feature space and the label space of the SD and the TD are same, but the marginal distribution ($P_s(\boldsymbol{x}^s) \neq P_t(\boldsymbol{x}^t)$) and the conditional distribution ($Q_s(\boldsymbol{y}^s|\boldsymbol{x}^s) \neq Q_t(\boldsymbol{y}^t|\boldsymbol{x}^t)$) are different. The experimental database contains C different classes and each class c has $n^{(c)}$ samples points in the subdomain $D^{(c)}$.

Given two points (i.e., feature vector) $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, the Mahalanobis distance can be described as

$$d_{\boldsymbol{M}}(\boldsymbol{x}_i - \boldsymbol{x}_j) = \sqrt{(\boldsymbol{x}_i - \boldsymbol{x}_j)^T \boldsymbol{M} (\boldsymbol{x}_i - \boldsymbol{x}_j)} \tag{1}$$

and parameterized by distance metric $\boldsymbol{M} \in \mathbb{S}_+^d$, where $\mathbb{S}_+^d$ is the cone of symmetric positive semidefinite (PSD) $d \times d$ real-valued matrix. $\boldsymbol{M} \in \mathbb{S}_+^d$ ensures that $d_{\boldsymbol{M}}$ satisfies the properties of a pseudo-distance, for example, identity, symmetry, nonnegativity, and triangle inequality [12]. When $\boldsymbol{M}$ is an identity matrix ($\boldsymbol{M} = \boldsymbol{I}$), $d_{\boldsymbol{M}}$ is recovered to the Euclidean distance. The matrix $\boldsymbol{M}$ can be decomposed into $\boldsymbol{M} = \boldsymbol{A}^T \boldsymbol{A}$, where $\boldsymbol{A}$ ($\boldsymbol{A} \geq 0$) is a transformation matrix.

Most methods learn the Mahalanobis distance metric in a weakly supervised way from PC including similar constraints (SC) $\mathscr{S}$ and dissimilar constraints (dc) $\mathscr{D}$ of the following form:

$$\begin{cases} \mathscr{S} = (\boldsymbol{x}_i, \boldsymbol{x}_j) : \boldsymbol{x}_i \text{ and } \boldsymbol{x}_j \text{ should be similar} \\ \mathscr{D} = (\boldsymbol{x}_i, \boldsymbol{x}_j) : \boldsymbol{x}_i \text{ and } \boldsymbol{x}_j \text{ should be dissimilar} \end{cases} \tag{2}$$

where the SC (DC) means that the pairwise sample points have the same (different) labels.

B. Three Optimization Components

In the process of handling the DS of SD and TD, TML uses the prior knowledge of SD to help TD learn distance metrics, but few methods can further take into account both DIP and GSP. In this article, we propose to learn the distance metrics by jointly optimizing three complementary optimization components, that is, PC, JDA, and MR, to achieve the DIP, GSP, and handle the SD simultaneously. More specifically, we preserve the discriminative information by introducing PC, align the marginal and conditional distribution by introducing JDA based on nonparametric maximum mean discrepancy (MMD), and explore the geometric structure properties of the neighbors by introducing MR into the final objective function. By this means, the proposed GTML can learn a robust distance metric with improved generalization.

1) Pairwise Constraint (PC): The primary goal of the TML method is to improve the discriminative ability of feature representations, thus it needs to learn a distance metric from a given collection of similar/dissimilar samples by punishing the large distances between similar pairs and the small distances between dissimilar pairs [15], [32]. The similar constraints loss (SCL) and dissimilar constraints loss (DCL) is denoted as $L_s(\cdot,\cdot)$ and $L_d(\cdot,\cdot)$, respectively

$$\begin{aligned} L_s(\boldsymbol{M}, \boldsymbol{S}) &= \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathscr{S})} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_{\boldsymbol{M}}^2 \\ &= \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathscr{S})} \mathrm{tr}((\boldsymbol{x}_i - \boldsymbol{x}_j)(\boldsymbol{x}_i - \boldsymbol{x}_j)^T \boldsymbol{M}) \\ &= \mathrm{tr}(\boldsymbol{S} \cdot \boldsymbol{M}) \end{aligned} \tag{3}$$

$$\begin{aligned} L_d(\boldsymbol{M}, \boldsymbol{D}) &= \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathscr{D})} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_{\boldsymbol{M}}^2 \\ &= \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathscr{D})} \mathrm{tr}((\boldsymbol{x}_i - \boldsymbol{x}_j)(\boldsymbol{x}_i - \boldsymbol{x}_j)^T \boldsymbol{M}) \\ &= \mathrm{tr}(\boldsymbol{D} \cdot \boldsymbol{M}) \end{aligned} \tag{4}$$

where

$$\begin{aligned} \boldsymbol{S} &= \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathscr{S})} (\boldsymbol{x}_i - \boldsymbol{x}_j)(\boldsymbol{x}_i - \boldsymbol{x}_j)^T \\ \boldsymbol{D} &= \sum_{(\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathscr{D})} (\boldsymbol{x}_i - \boldsymbol{x}_j)(\boldsymbol{x}_i - \boldsymbol{x}_j)^T. \end{aligned} \tag{5}$$

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2) Joint Distribution Adaptation (JDA): Since learning distance metric based on PC requires the training and test data to be sampled from identical probability distribution, thus it needs to minimize the difference between the joint probability distributions [17], [66], that is, marginal distribution and conditional distribution as shown in Fig. 2. In the implementation, the nonparametric maximum mean discrepancy (MMD) is used as the distribution distance measure. The JDA is denoted as

$$R_d(\boldsymbol{M}, \boldsymbol{L_{md}}, \boldsymbol{L_{cd}}) = R_{md}(\boldsymbol{M}, \boldsymbol{L_{md}}) + R_{cd}(\boldsymbol{M}, \boldsymbol{L_{cd}}) \quad (6)$$

where the $R_{md}(\cdot,\cdot)$ and $R_{cd}(\cdot,\cdot)$ denotes the MDA and CDA, respectively

$$\begin{aligned} R_{md}(\boldsymbol{M}, \boldsymbol{L_{md}}) &= \|\frac{1}{n_s}\sum_{i=1}^{n_s} \boldsymbol{A}\boldsymbol{x}_i^s - \frac{1}{n_t}\sum_{j=1}^{n_t} \boldsymbol{A}\boldsymbol{x}_j^t\|^2 \\ &= \mathrm{tr}(\boldsymbol{X}\boldsymbol{L_{md}}\boldsymbol{X}^T\boldsymbol{M}) \quad (7) \\ R_{cd}(\boldsymbol{M}, \boldsymbol{L_{cd}}) &= \sum_{c=1}^{C} \|\frac{1}{n_s^{(c)}} \sum_{\boldsymbol{x}_i \in D_s^{(c)}} \boldsymbol{A}\boldsymbol{x}_i^s - \frac{1}{n_t^{(c)}} \sum_{\boldsymbol{x}_j \in D_t^{(c)}} \boldsymbol{A}\boldsymbol{x}_j^t\|^2 \\ &= \sum_{c=1}^{C} \mathrm{tr}(\boldsymbol{X}\boldsymbol{L_{cd}}\boldsymbol{X}^T\boldsymbol{M}) \quad (8) \end{aligned}$$

where $\boldsymbol{X} = \{\boldsymbol{x}_1^s, \ldots, \boldsymbol{x}_{n_s}^s, \boldsymbol{x}_1^t, \ldots, \boldsymbol{x}_{n_t}^t\} \in \mathbb{R}^{d\times(n_s+n_t)}$

$$(\boldsymbol{L_{md}})_{ij} = \begin{cases} \dfrac{1}{n_s n_s}, & \boldsymbol{x_i}, \boldsymbol{x_j} \in D_s \\ \dfrac{1}{n_t n_t}, & \boldsymbol{x_i}, \boldsymbol{x_j} \in D_t \\ \dfrac{-1}{n_s n_t}, & \text{otherwise} \end{cases} \quad (9)$$

$$(\boldsymbol{L_{cd}})_{ij} = \begin{cases} \dfrac{1}{n_s^{(c)} n_s^{(c)}} & \text{if } \boldsymbol{x_i}, \boldsymbol{x_j} \in D_s^{(c)} \\ \dfrac{1}{n_t^{(c)} n_t^{(c)}} & \text{if } \boldsymbol{x_i}, \boldsymbol{x_j} \in D_t^{(c)} \\ \dfrac{-1}{n_s^{(c)} n_t^{(c)}}, & \begin{cases} \boldsymbol{x_i} \in D_s^{(c)}, \boldsymbol{x_j} \in D_t^{(c)} \\ \boldsymbol{x_i} \in D_t^{(c)}, \boldsymbol{x_j} \in D_s^{(c)}. \end{cases} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

3) Manifold Regularization (MR): Additionally, we add a Laplacian regularization term to further exploit the similar geometric structure property of nearest points [15], [66]. As a regularization, MR helps for preventing overfitting and improves the smoothness, robustness, and generalization performance of the distance metrics [32]. The MR is denoted as

$$\begin{aligned} R_m(\boldsymbol{M}, \boldsymbol{L_m}) &= \frac{1}{2}\sum_{i,j=1} \|\boldsymbol{A}\boldsymbol{x_i} - \boldsymbol{A}\boldsymbol{x_j}\|^2 \boldsymbol{W_{ij}} \\ &= \sum_{l=1}^{d} \boldsymbol{a_l}^T \boldsymbol{X}(\boldsymbol{G} - \boldsymbol{W_{ij}})\boldsymbol{X}^T \boldsymbol{a_l} \\ &= \mathrm{tr}(\boldsymbol{X}\boldsymbol{L_m}\boldsymbol{X}^T\boldsymbol{M}) \quad (11) \end{aligned}$$

where $\boldsymbol{W_{ij}}$ is the graph affinity matrix between $\boldsymbol{x_i}$ and $\boldsymbol{x_j}$, $\boldsymbol{L_m} = \boldsymbol{G} - \boldsymbol{W_{ij}}$ is the graph Laplacian matrix, $\boldsymbol{G}$ is a diagonal matrix whose diagonal elements are equal to the summation of the row entries of $\boldsymbol{W_{ij}}$, that is, $\boldsymbol{G_{ii}} = \sum_j \boldsymbol{W_{ij}}$, $\boldsymbol{a_l}$ is the lth column of matrix $\boldsymbol{A}$, $\boldsymbol{W_{ij}}$ is defined as

$$\boldsymbol{W_{ij}} = \begin{cases} 1, & \boldsymbol{x_i} \in N(\boldsymbol{x_j}) \text{ or } \boldsymbol{x_j} \in N(\boldsymbol{x_i}) \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

where $N(\boldsymbol{x_j})$ denotes the k-NN list of the data point $\boldsymbol{x_j}$.

C. Two Optimization Strategies

In practical applications, there are usually two ways to fuse SCL and DCL (see Section III-B) to construct MML, which are referred as the accumulative MML (A-MML)

$$L(\boldsymbol{M}, \boldsymbol{S}, \boldsymbol{D}) = L_s(\boldsymbol{M}, \boldsymbol{S}) - L_d(\boldsymbol{M}, \boldsymbol{D}) \quad (13)$$

and the ratio MML (R-MML)

$$L(\boldsymbol{M}, \boldsymbol{S}, \boldsymbol{D}) = \frac{L_s(\boldsymbol{M}, \boldsymbol{S})}{L_d(\boldsymbol{M}, \boldsymbol{D})}. \quad (14)$$

According to these two variations, we can construct two different objective functions which termed as GTML-A which utilizes A-MML and GTML-R which utilizes R-MML in this article.

GTML-A:

$$\arg\min_{\boldsymbol{M}\geq 0} \ \eta_s L_s - \eta_d L_d + \mu_m R_{md} + \mu_c R_{cd} + \gamma R_m \quad (15)$$

GTML-R:

$$\arg\min_{\boldsymbol{M}\geq 0} \ \frac{\eta_s L_s}{\eta_d L_d} + \mu_m R_{md} + \mu_c R_{cd} + \gamma R_m \quad (16)$$

where η_s, η_d, μ_m, μ_c and γ are the positive hyperparameters. The optimal distance metric $\boldsymbol{M^*}$ is learned when objective function is minimized, which is equivalent to minimize the R_{md}, R_{cd}, R_m, and $L_s - L_d$ for GTML-A (or $\frac{\eta_s L_s}{\eta_d L_d}$ for GTML-R). We notice that the formula (16) can be well established as a convex optimization function by enforcing an equality constraint as $\eta_d L_d = 1$, thus it can be rewritten as

GTML-R:

$$\begin{aligned} &\arg\min_{\boldsymbol{M}\geq 0} \ \eta_s L_s + \mu_m R_{md} + \mu_c R_{cd} + \gamma R_m \\ &\text{s.t.} \quad \eta_d L_d = 1. \end{aligned} \quad (17)$$

The objective function (15) and (17) are clearly standard formulations of semidefinite programs (SDP), which can be solved efficiently with global optimum using existing convex optimization packages, such as SDPT3 in CVX toolbox [75], when all hyperparameters are fixed.

D. Solutions for ZLS and SLS Tasks

In many real-world application scenarios, there are two typical tasks, that is, 1) the task of ZLS in TD and 2) the task of SLS in TD. For ZLS task, the corresponding applicable solution is shown in Fig. 3, since there is no labeled sample available in TD, the proposed GTML first apply labeled samples in SD to learn a pseudo-TD classifier, then uses it to classify unlabeled TD data to obtain pseudo-labels. These pseudo-labeled TD data along with labeled SD data are then used to construct JDA. Both labeled SD data and unlabeled

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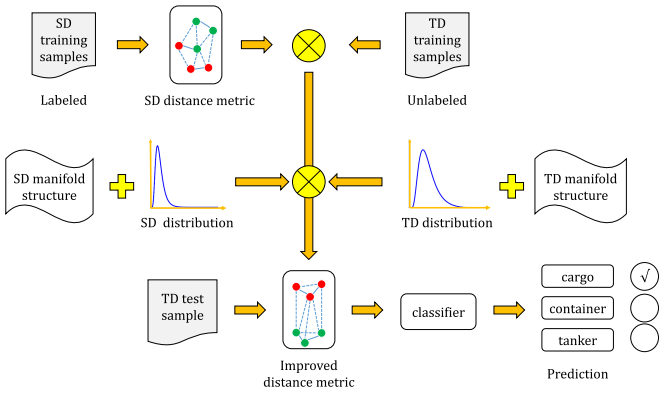


Fig. 3. Solution of GTML for ZLS.

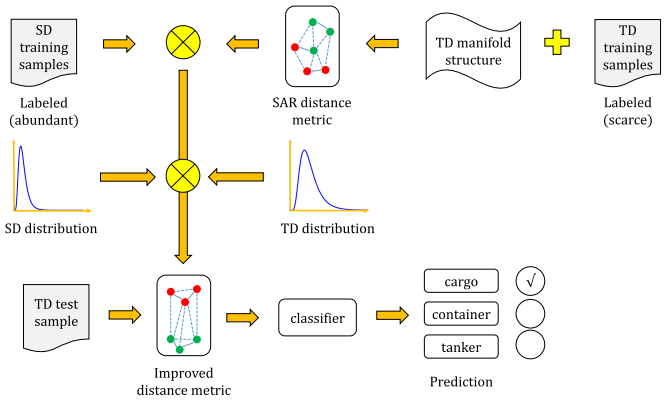


Fig. 4. Solution of GTML for SLS.

Algorithm 1 GTML for ZLS

Require: Input SD labeled samples $\boldsymbol{X}_l^s$ and corresponding labels $\boldsymbol{Y}^s$, TD unlabeled data $\boldsymbol{X}_u^t$, the hyperparameters η_s, η_d, μ_m, μ_c, γ and neighbor k;

Ensure: Distance metric $\boldsymbol{M}^*$

1: Initializing distance metric $\boldsymbol{M}$ based on identity matrix $\boldsymbol{I}$.
2: Training a k-NN classifier with $\boldsymbol{X}_l^s$ and $\boldsymbol{Y}^s$ to obtain pseudo-labels $\hat{\boldsymbol{Y}}^t$ of $\boldsymbol{X}_u^t$.
3: Constructing SC $\mathscr{S}$ and DC $\mathscr{D}$ according to the formula (2) based on $\boldsymbol{X}_l^s$ and $\boldsymbol{Y}^s$.
4: Constructing SCL L_s according to the formula (3) based on SC; Constructing DCL L_d according to the formula (4) based on DC.
5: Constructing MDA L_{md} according to the formula (7) based on $\boldsymbol{X}_l^s$ and $\boldsymbol{X}_u^t$; Constructing CDA L_{cd} according to the formula (8) based on $\boldsymbol{X}_l^s$, $\boldsymbol{Y}^s$, $\boldsymbol{X}_u^t$, and $\hat{\boldsymbol{Y}}^t$.
6: Constructing MR L_m according to the formula (11) based on $\boldsymbol{X}_l^s$ and $\boldsymbol{X}_u^t$.
7: Return distance metric $\boldsymbol{M}^*$ according to objection function (15) or (17) based on CVX convex optimization toolbox.

Algorithm 2 GTML for SLS

Require: Input SD labeled samples $\boldsymbol{X}_l^s$ and corresponding labels $\boldsymbol{Y}^s$, TD labeled samples $\boldsymbol{X}_l^t$ and corresponding labels $\boldsymbol{Y}^t$, the hyperparameters η_s, η_d, μ_m, μ_c, γ, and neighbor k;

Ensure: Distance metric $\boldsymbol{M}^*$

1: Initializing distance metric $\boldsymbol{M}$ based on identity matrix $\boldsymbol{I}$.
2: Constructing SC $\mathscr{S}$ and DC $\mathscr{D}$ according to the formula (2) based on $\boldsymbol{X}_l^t$ and $\boldsymbol{Y}^t$.
3: Constructing SCL L_s according to the formula (3) based on SC; Constructing DCL L_d according to the formula (4) based on DC.
4: Constructing MDA L_{md} according to the formula (7) based on $\boldsymbol{X}_l^s$ and $\boldsymbol{X}_l^t$; Constructing CDA L_{cd} according to the formula (8) based on $\boldsymbol{X}_l^s$, $\boldsymbol{Y}^s$, $\boldsymbol{X}_l^t$, and $\boldsymbol{Y}^t$.
5: Constructing MR L_m according to the formula (11) based on $\boldsymbol{X}_l^t$.
6: Return distance metric $\boldsymbol{M}^*$ according to objection function (15) or (17) based on CVX convex optimization toolbox.

TD data are adopted to construct MR. The labeled SD data is applied to constructed PC. The detailed pseudo-code is described in Algorithm 1. For SLS task, the corresponding applicable solution is shown in Fig. 4. GTML adopts labeled data in TD to construct PC and MR, while construct JDA based on the data in both domains. The detailed pseudo-code is described in Algorithm 2.

IV. Experiments and Discussion

A. Comparison Methods

In this section, we compared the performance of the proposed GTML with several state-of-the-art methods on both ZLS and SLS tasks. For ZLS task, since there is no labeled sample available in TD, DML methods are not feasible for this task, thus we compared fifteen TL methods and three TML methods with GTML. A brief description of the optimization component for each method and its corresponding document is listed in Table I. For SLS task, we compared GTML with sixteen DML methods, four TL methods, and three TML methods. A brief description of the optimization component for each method and its corresponding document is listed in Table II.

B. Data Preparation and Experiment Protocol

1) Data: In the comparative experiments, we investigated the performance of GTML on the task of ship classification in SAR images (TD) by the assistance of AIS data (SD). *AIS* followed literature [16], which has 600 ship samples per class. There are two SAR ship data sets followed literature [13]. *HR-SAR* is a high-resolution SAR ship data set which was collected by Xing *et al.* [4] from six TerraSAR-X stripmap-mode SAR images with 2.0 m × 1.5 m resolution in azimuth and range directions, which has 50 ship samples per class. *MR-SAR* is a moderate-resolution SAR ship data set which was collected by Lang *et al.* [5], [8] from eight Radarsat-2 standard-mode VV-polarization images with moderate resolution of about 15 m both in azimuth and range directions, which has 50 ship samples per class. This study assumes that SD and TD have the same classification task,

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TABLE I
COMPARISON METHODS FOR ZLS TASK

	Method	Optimization			Ref.
		DIP	DA	GSP	
TL	STL		CDA		[45]
	TCA		MDA		[46]
	TJM		MDA · IR		[49]
	TLR		MDA	RC	[52]
	JDA		JDA		[24]
	VDA	WSM	JDA		[53]
	BDA		JDA		[54]
	JGSA	WSM · ISM	JDA · SDM	TVM	[55]
	DLC-DA	LCR · IRF	JDA · LCR		[56]
	W-JDA		JDA · IR		[57]
	GFK		GFK		[58]
	LDADA		WSM · ISM		[62]
	LSDT		RC	Sc · RC	[63]
	LRSR	LCR	RC · LCR	Sc · LC	[64]
	LRDR	LCR · MC · LE	RC · LCR	Sc · LC	[65]
TML	ITL	MI	MI		[73]
	CDML	MR	MDA	MR	[72]
	RTML		JDA	LC · RC	[23]

TABLE II
COMPARISON METHODS FOR SLS TASK

	Method	Optimization			Ref.
		DIP	DA	GSP	
DML	MMC	PC · MML			[29]
	TrDCA	PC · MML		LC	[31]
	LRML	PC · MML		MR	[32]
	DML-eig	PC · MML		LC	[33]
	RDML	PC · HL		FN · LC	[34]
	SML	PC · HL		Sc · LC	[35]
	Sub-SML	PC · HL		FN	[36]
	D-SVM	PC · HL		FN	[37]
	LMNN	TC · HL			[38]
	T-SVM	TC · HL		FN	[37]
	LODML	TC · HL		LTC	[37]
	BoostMetric	PC · EL		LC	[40]
	DML-dc	PC · RL · LMP		LC	[41]
	ITML	PC		LD · LC	[42]
	DMLMJ	PC · JD		LC	[43]
	NCA	PC · LOE		LC	[44]
TL	DAMA	MR	MR	MR	[68]
	MMDT	HL	SA	FN	[69]
	HFA	HL	SA	FN	[70]
	CDLS		JDA	L_2	[71]
TML	L-DML	PC · MML	LD	LD	[27]
	M-DML	PC · MML	MR	MR	[27]
	MTLF	PC · MML	IR · MML	LC	[26]

∗ Here we just apply the MMDT, HFA, and CDLS methods themselves. The features used are still *homogeneous*, that is, in the SD and TD domains, we use the same features.

therefore above database contains three same ship classes, that is, cargos, containers, and oil tankers.

As aforementioned, the comparison experiments were conducted on two different tasks, that is, ZLS task and SLS task. For ZLS task, we randomly selected 300 labeled data per class in SD and unlabeled data[3] in TD to train the classifier, then evaluated the performance of learned classifier in TD. For SLS task, we randomly selected 300 labeled data per class in SD and three labeled data per class in TD to train classifier, then evaluated the performance of learned classifier on remaining data in TD.

[3]Since ZLS task focuses on the case that there is no labeled data available in TD, thus we deliberately do not use the labels of TD data.

2) Feature: Considering that the AIS can only provide the length and width of the ship as the transferable features, and this study focuses on the *homogeneous* TML problem, thus we still followed [16] and extracted the NGFs as the ship representation. In this way, the ship is represented as a vector which is a concatenation of the features $\upsilon = [f_1, f_2, \ldots, f_{10}, f_{11}]$, where f_1, f_2, f_3, ... represents length, width, perimeter, ..., respectively.

3) Classifier: The purpose of this study is not to obtain higher SAR ship classification accuracy in the two tasks of ZLS and SLS, but only to prove the advantages of the proposed GTML over the existing methods in improving the SAR ship classification performance. Therefore, we chose the 1-NN classifier as the base classifier for all methods, except MMDT, CDLS, and HFA, which use the SVM based on radial basis function (RBF) as the base classifier following the suggestion of the original literature. Considering that AIS→SAR is a completely new task for all methods, to ensure that each method achieves its optimal performance, we use the grid search method to find the optimal result of each method on test data. For the proposed GTML, the optimal parameter corresponding to each optimization component was searched in a universal search space $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. For the comparison methods, we refer to the universal search space to extend the suggested parameter search space by the original literature to a wider scope.

4) Evaluation: To gain a statistically meaningful understanding for each method, the experiment was conducted ten times to eliminate the possible bias caused by random sampling to data in each individual experiment. The mean and standard deviation of acc, which is defined as [15], [17], are used as the evaluation metric

$$acc = \frac{|\{\boldsymbol{x} : \boldsymbol{x} \in D_t \wedge f(\boldsymbol{x}) = y(\boldsymbol{x})\}|}{|\{\boldsymbol{x} : \boldsymbol{x} \in D_t\}|} \tag{18}$$

where $y(\boldsymbol{x})$ is the ground truth label of $\boldsymbol{x}$ while $f(\boldsymbol{x})$ is the label predicted by the classification algorithm. We also reported the mean performance (MP) on HR-SAR and MR-SAR data sets, to present overall performance of comparative methods.

C. Results and Discussion of ZLS Task

1) Classification Performance: The average classification accuracy of 10 experiments for each method are illustrated in Table III. For HR-SAR data set, the best average classification accuracy is 75.33% obtained by GTML-R, outperforms the second best 74.27% (JGSA) by 1.06%, the third best 67.47% (GTML-A) by about 8.00%, and all other methods by more than 10.00%. For MR-SAR data set, due to the lower resolution of SAR images compared to HR-SAR, the performance of all methods drops significantly. The best classification accuracy is only 61.53% obtained by JGSA, followed closely by 60.40% and 59.73% obtained by the proposed GTML-A and GTML-R, respectively. When comparing MP which reflects the overall performance of various methods on two data sets, it shows that the proposed GTML-R (67.53%) and JGSA (67.90%) perform significantly better than other methods. They are better than GTML-A (63.94%) which is

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TABLE III
MEAN AND STANDARD DEVIATION OF ACCURACY IN ZLS (%)

	1-NNs	STL	TCA	TJM	TLR	JDA	VDA	BDA
AIS→HR-SAR	41.33±5.96	38.33±4.11	54.20±4.17	52.47±2.76	42.47±7.81	56.87±3.84	52.47±4.49	59.47±3.99
AIS→MR-SAR	48.07±4.17	44.13±2.75	51.07±2.31	47.53±3.51	43.20±5.58	52.60±2.19	52.93±3.10	54.67±2.83
MP	44.70	41.23	52.64	50.00	42.84	54.74	52.70	57.07
runtime/s	0.0081	0.3290	0.0317	0.0879	0.0373	0.0839	1.1973	0.0843
	JGSA	DLC-DA	W-JDA	GFK	LDADA	LSDT	LRSR	LRDR
AIS→HR-SAR	74.27±3.34	58.33±5.61	57.87±3.99	40.67±6.09	45.93±0.86	48.13±6.26	64.20±6.05	64.27±6.22
AIS→MR-SAR	61.53±3.77	55.87±3.59	52.80±2.08	49.73±4.11	56.00±0.94	44.60±2.48	57.33±5.45	57.07±5.55
MP	67.90	57.10	55.34	45.20	50.97	46.37	60.77	60.67
runtime/s	0.5503	0.4079	0.0761	6.2066	0.0338	17.4791	2.9543	12.2902
	ITL	CDML	RTML	GTML-A	GTML-R			
AIS→HR-SAR	53.53±4.10	39.20±4.65	61.40±2.36	67.47±4.21	75.33±5.89			
AIS→MR-SAR	52.07±1.87	41.40±4.09	54.20±4.75	60.40±2.87	59.73±4.52			
MP	52.80	40.30	57.80	63.94	67.53			
runtime/s	11.7166	3.1767	3.5109	0.4288	0.4514			

the second best more than 3.00%, and the third best method LRSR (60.77%) and LRDR (60.67%) close to 7.00%. The experimental results prove the effectiveness of the proposed GTML, and also illustrate that GTML-R which applies the R-MML strategy is more suitable for ZLS task.

2) Component Evaluation: Analyzing the objective functions and optimization components of various methods as listed in Table I, we find that the above results are theoretically reasonable. It is worth noting that among the TL methods, STL, TCA, JDA, etc. only consider DA but ignore DIP and GSP. VDA, DLC-DA, LSDT, etc. consider two of the three, but not all. It can be inferred that it is the reason that the MP of these methods does not reach 60.00%. A similar situation occurs with three TML methods, that is, ITL, CDML, and RTML. Although these methods combine TL and DML, they do not get good results because three key optimization components are not fully considered. As aforementioned, RTML can effectively mitigate the DS, but neglect the importance of DIP. In contrast, LRDR, LRSR, and JGSA take both DIP, DA, and GSP into consideration, therefore the performance of these methods has significantly improved, LRDR and LRSR are close to 61.00%, and JGSA reaches even to 67.90%. As surveyed in Section II, JGSA considers more optimization components, it performs JDA and SDM to reduce the discrepancy between SD and TD, leverages WSM and ISM to achieve DIP, and introduces TVM to achieve GSP. In contrast, the proposed GTML employs a simpler but effective optimization strategy to implement DIP, DA, and GSP as described in Section III-B, but achieves the same performance as JGSA. Experiments demonstrate that the proposed optimization components are well complementary, and can effectively solve the DS problem while learning a highly discriminative distance metric, thereby improving the ship classification accuracy in SAR domain.

3) Ablation Study: To inspect the effectiveness of a specific optimization component in (15) and (17) to the classification performance, we conducted ablation study. This time, we run GTML again, but removed one of the five optimization components, that is, SCL, DCL, MDA, CDA, and MR, in sequence, and recorded the decline in classification accuracy. Table IV illustrates the classification accuracy and corresponding decline after an optimization component removal.

TABLE IV
PERFORMANCE DECLINE AFTER REMOVING A OPTIMIZATION COMPONENT IN ZLS (%)

	GTML-A	-SCL	-DCL	-MDA	-CDA	-MR
HR-SAR	67.47	65.07	60.13	62.73	63.27	58.07
Decline	/	-1.21	-1.77	-2.06	-1.28	-5.18
MR-SAR	60.40	58.60	56.13	54.40	58.40	56.20
Decline	/	-1.80	-4.27	-6.00	-2.00	-4.20
	GTML-R	-SCL	-DCL	-MDA	-CDA	-MR
HR-SAR	75.33	58.07	60.13	71.93	74.47	66.20
Decline	/	-17.26	-15.20	-3.40	-0.89	-9.13
MR-SAR	59.73	58.67	56.13	55.87	57.47	49.07
Decline	/	-1.06	-3.60	-3.86	-2.26	-10.66

It can be seen that the removal of each optimization component leads to a certain degree of degradation in classification performance. It shows that the five optimization components have strong complementarity and jointly contribute to the improvement of classification performance. Because GTML-A and GTML-R use different optimization components, GTML-A uses the A-MML and GTML-R uses the R-MML, resulting in different effects of SCL and DCL on the classification performance. For instance, in the HR-SAR data set, for GTML-A, when SCL and DCL are removed, the accuracy decreases by 1.21% and 1.77%, respectively. However, for GTML-R, this effect is more serious, the removal of SCL and DCL causes a drop in accuracy by as much as 17.26% and 15.20%, respectively. While, for MR-SAR data, the removal of two optimization components has little effect on both GTML-A and GTML-R, which implies that the low- and moderate-resolution SAR images can only provide limited discriminative information, so DIP is more effective for high-discrimination data than for low-discrimination data. The experimental results highlight that the proposed GSP, which is implemented by MR, has a significant impact on classification performance: For HR-SAR, the removal of MR leads to the accuracy decline up to 5.18% for GTML-A and 9.13% for GTML-R, respectively. For MR-SAR, the corresponding accuracy drops by up to 4.20% for GTML-A and 10.66% for GTML-R, respectively.

4) Feature Visualization: Above analysis can be verified again from the Fig. 5, which illustrates the feature distribution in both domains after GTML-R. For HR-SAR data set,

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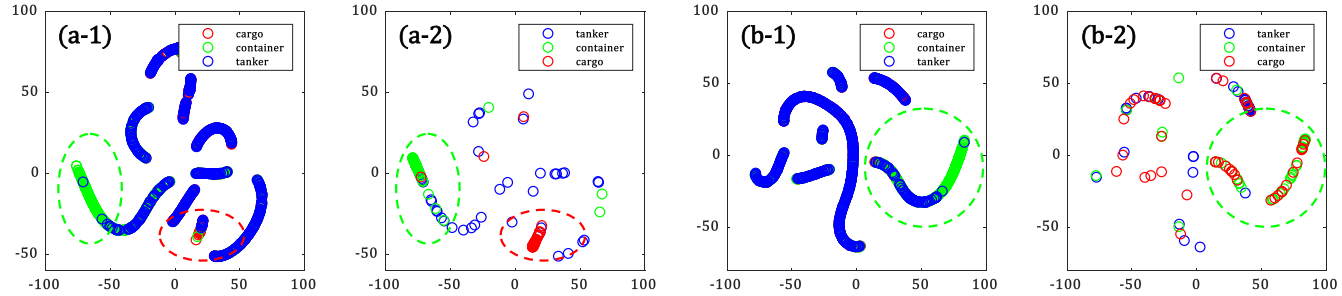


Fig. 5. Visualization of ship samples scattered in the feature space which is reduced to a 2-D perspective based on t-SNE method [3]. For ZLS task, subplot (a-1) and (a-2) represent the scatter of AIS domain and HR-SAR domain after GTML-R ($\boldsymbol{XM}$), respectively. Subplot (b-1) and (b-2) represent the scatter of AIS domain and MR-SAR domain after GTML-R ($\boldsymbol{XM}$), respectively.

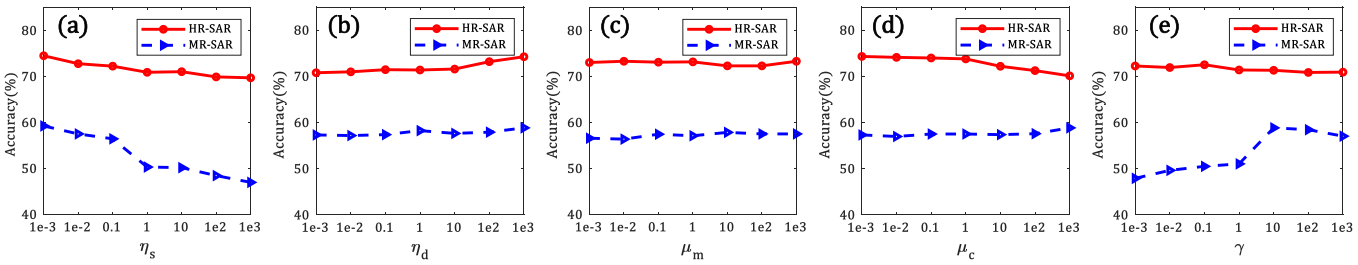


Fig. 6. Parameter sensitivity analysis of GTML-R on the ZLS task. (a)–(e) represents the classification performance (acc) varies with the parameter (a) η_s, (b) η_d, (c) μ_m, (d) μ_c, and (e) γ, respectively.

comparing Fig. 5(a-1) and (a-2), and comparing this figure with Fig. 2, it shows that both the marginal distribution and conditional distribution are well aligned. Due to the usage of GSP, the geometric structure of AIS domain and SAR domain has good consistency, which can be reflected in the parts marked by the red and green ovals in the figure. Although no labeled samples are available in the SAR domain (TD) in ZLS task, thanks to the use of PC, with the help of labeled samples in the AIS domain, the discriminative information is well transferred to the SAR domain, so that the subordinate-level categories are well distinguished. It can be seen from Fig. 5(a-1) and (a-2) that the vast majority of "cargos" are concentrated in the red ellipse region, while most of "containers" are concentrated in the green ellipse region on the left. For MR-SAR data set, as shown in Fig. 5(b-1) and (b-2), although the marginal distributions are aligned, due to the complexity and challenges of MR-SAR data set, there remains serious confusion between different subordinate-level categories, making it difficult to align the conditional distributions. Nonetheless, as aforementioned, since GSP is implemented using MR, most of samples belonging to "containers" category are still clustered together in the green ellipse region on the right side of Fig. 5(b-2).

5) Confusion Matrix: As shown in Table V, the corresponding confusion matrix of GTML-R on ZLS task also presents consistent result with above analysis. In HR-SAR domain, the "cargo" class is clearly separated from "container" class and "tanker" class, thus it gets a *recall* up to 94.20%. In contrast, since "tanker" class is not clearly distinguishable from "cargo" class and "container" class, there is still a certain degree of mixing, its *recall* is only 44.20%. In MR-SAR domain, the "container" class are well distinguished from other classes, thus it gets higher *recall* 78.20% than "cargo" class and "tanker" class.

TABLE V
CONFUSION MATRIX OF GTML-R IN ZLS (%)

HR-SAR	cargo	container	tanker	*recall*
cargo	471	13	16	94.20%
container	21	438	41	87.60%
tanker	117	162	221	44.20%
precision	77.34%	71.45%	79.50%	75.33%
MR-SAR	cargo	container	tanker	*recall*
cargo	226	196	78	45.20%
container	73	391	36	78.20%
tanker	166	55	279	55.80%
precision	48.60%	60.90%	70.99%	59.73%

6) Parameter Sensitivity: As described in Section III, the proposed GTML involves five parameters. Theoretically, larger values of η_s and η_d make DIP more important, larger values of μ_m and μ_c make DA more effective, and larger value of γ makes GSP more important. We conduct empirical parameter sensitivity analysis through running GTML (GTML-R in this experiment) with a wide range of parameter values ($\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$) on both AIS→HR-SAR/MR-SAR tasks. The classification accuracy w.r.t. different values of η_s, η_d, μ_m, μ_c, and γ are plotted in Fig. 6(a)–(e), respectively. Observing Fig. 6(a) and (b), it shows that the classification performance becomes a little better as η_d increases, but becomes a little worse as η_s increases. This shows that DCL (η_d) is more important than SCL (η_s) in terms of DIP. Fig. 6(c)–(e) shows that MDA (μ_m) and CDA (μ_c) maintain a stable state in terms of DA, and a

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TABLE VI
MEAN AND STANDARD DEVIATION OF ACCURACY IN SLS (%)

	Eu-1-NNt	MMC	TrDCA	LRML	DML_eig	RDML	SML	Sub-SML	D-SVM
HR-SAR	75.67±5.27	79.29±7.26	71.84±5.67	76.45±7.26	78.01±5.08	76.74±6.58	80.87±6.56	75.67±5.27	77.38±4.48
MR-SAR	58.65±7.92	61.28±8.77	63.76±6.38	63.33±6.74	62.84±7.29	54.96±10.67	63.90±6.72	59.22±7.29	61.28±6.18
MP	67.16	70.29	67.80	69.89	70.43	65.85	72.39	67.45	69.33
runtime/s	0.0027	0.0044	0.0040	0.3800	0.0301	0.0052	0.1246	0.1301	0.0050
	LMNN	T-SVM	LODML	BoostMetric	DML-dc	ITML	DMLMJ	NCA	
HR-SAR	77.16±4.36	76.88±6.04	79.22±7.28	78.58±10.16	75.82±5.25	79.79±6.46	72.91±9.29	75.67±5.27	
MR-SAR	59.15±7.69	60.92±8.21	59.43±7.59	58.51±10.12	59.72±7.21	62.34±6.57	59.72±8.86	58.65±7.92	
MP	68.16	68.90	69.33	68.55	67.77	71.07	66.32	67.16	
runtime/s	0.0398	0.0035	0.0815	0.5501	0.1215	0.0977	0.0112	0.0052	
	DAMA	MMDT	HFA	CDLS	L-DML	M-DML	MTLF	GTML-A	GTML-R
AIS→HR-SAR	77.16±7.85	76.24±4.44	76.95±4.45	75.74±3.50	77.66±6.25	79.01±6.69	76.10±4.96	84.26±3.22	83.76±4.45
AIS→MR-SAR	50.14±8.05	59.22±4.04	67.16±3.68	65.82±3.13	64.11±6.17	63.69±6.42	60.14±7.31	68.01±4.39	66.88±4.50
MP	63.65	67.73	72.06	70.78	70.89	71.35	68.12	76.14	75.32
runtime/s	1.8565	0.1945	0.5097	0.0768	1.7499	1.5310	0.0292	0.3519	0.3804

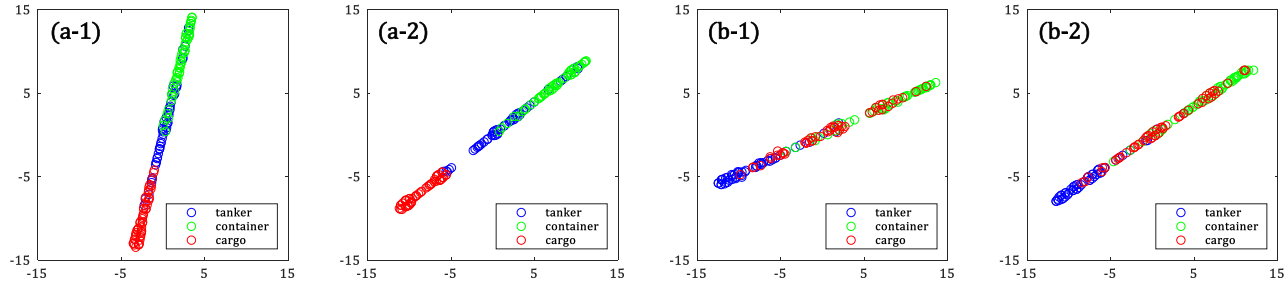


Fig. 7. Visualization of ship samples scattered in the feature space which is reduced to a 2-D perspective based on t-SNE method [3]. For SLS task, subplot (a-1) and (a-2) represent the original scatter of HR-SAR, and the scatter after GTML-A ($\boldsymbol{XM}$), respectively. Subplot (b-1) and (b-2) represent the original scatter of MR-SAR, and the scatter after GTML-A ($\boldsymbol{XM}$), respectively.

relatively large MR (γ) helps improve model performance. The results also show that, in contrast, the parameters are more sensitive to AIS→MR-SAR task, because MR-SAR is more complex than HR-SAR as aforementioned. To sum up, the performance of GTML stays robust with a wide range of regularization parameter choice.

7) Time Complexity: We empirically check the time complexity of all algorithms by running them on the AIS→HR-SAR/MR-SAR tasks.[4] The environment was an Intel Core i7-6800K CPU with 64 GB memory. The results in Table III reveal that except its superiority in classification accuracy, GTML also achieved a running time complexity better than other well-performing methods like JGSA, LRSR, LRDR whose MP exceed 60.00%.

D. Results and Discussion of SLS Task

1) Classification Performance: The results of comparative experiments on SLS tasks are listed in the Table VI. For HR-SAR data, the leading classification performance are obtained by the two GTML methods, GTML-A (84.26%) and GTML-R (83.76%), respectively. This result is far higher than what they achieved in the ZLS task. It is not surprising because unlike the ZLS task, there are labeled samples in SAR domain available in the SLS task, and these labeled samples play a vital and positive role in learning a good classifier. When labeled samples are available in TD, even if the number is small (e.g., three in this study), the supervised learning-based DML methods can effectively improve the fine-grained classification performance, only relies on learning a better distance metric to enhance the discriminability between subordinate-level categories, without considering domain adaption problem. It shows that the performance of SML (80.87%), ITML (79.79%), MMC (79.29%), and LODML (79.22%) is only lower than the proposed GTML, but higher than the best TML method (M-DML: 79.01%) and the best TL method (DAMA: 77.16%). On the other hand, for MR-SAR data, the DML methods have not achieved better results than the TL and TML methods. No DML method has a performance higher than 64.00%. In contrast, the two TL methods HFA and CDLS have reached 67.16% and 65.82%, respectively, the existing TML method L-DML has reached 64.11%, and the proposed GTML-A and GTML-R have reached 68.12% and 66.88%, respectively. This result proves that when the features (NGFs used in this study) extracted from the SAR images with relatively low resolution are not enough for the DML methods to learn a good distance metric, the important role of TL methods, especially the TML methods, that transfer other domain information to assist in the classification of the TD, is revealed.

2) Feature Visualization: We investigated the discriminative ability of original NGFs and the feature representations after GTML as shown in Fig. 7. This figure illustrates the scatter gram of ship samples in a 2-D feature space which is visualized based on t-SNE method [3]. Comparing Fig. 7(a-1) and (a-2), we can see that the intraclass tightness has been significantly enhanced, and GTML-A greatly reduces the mixing between "cargo" and "tanker" on HR-SAR data.

[4]Since HR-SAR and MR-SAR databases have the same sample number and feature dimension, the time cost is same for a given method.

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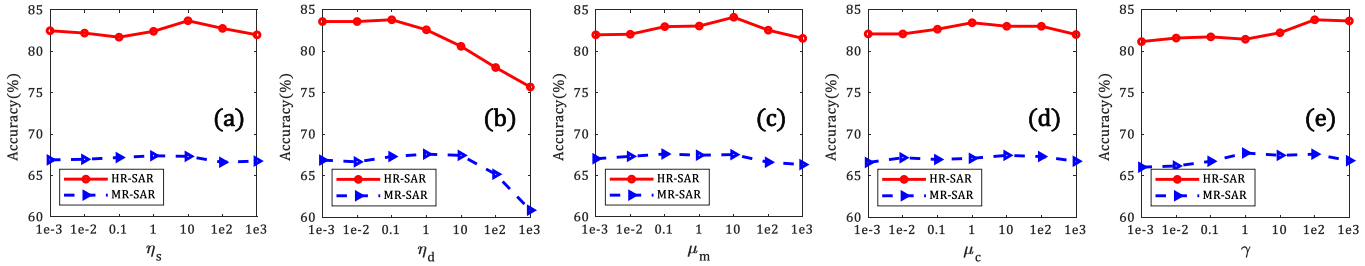


Fig. 8. Parameter sensitivity analysis of GTML-A on the SLS task. (a)–(e) represents the classification performance (acc) varies with the parameter (a) η_s, (b) η_d, (c) μ_m, (d) μ_c, and (e) γ, respectively.

TABLE VII
CONFUSION MATRIX OF GTML-A IN SLS (%)

HR-SAR	cargo	container	tanker	*recall*
cargo	445	4	21	94.68%
container	0	411	59	87.45%
tanker	28	110	332	70.64%
precision	94.08%	78.29%	80.58%	84.26%
MR-SAR	cargo	container	tanker	*recall*
cargo	257	161	52	54.68%
container	124	342	4	72.77%
tanker	91	19	360	76.60%
precision	54.45%	65.52%	86.54%	68.01%

TABLE VIII
PERFORMANCE DECLINE AFTER REMOVING A OPTIMIZATION COMPONENT IN SLS.(%)

	GTML-A	-SCL	-DCL	-MDA	-CDA	-MR
HR-SAR	84.26	82.20	82.84	80.71	81.63	81.49
Decline	/	-2.06	-1.42	-3.55	-2.63	-2.77
MR-SAR	68.01	66.95	65.11	66.10	66.74	65.89
Decline	/	-1.06	-2.90	-1.91	-1.27	-2.12
	GTML-R	-SCL	-DCL	-MDA	-CDA	-MR
HR-SAR	83.76	81.35	82.84	81.06	82.91	82.06
Decline	/	-2.41	-0.92	-2.70	-0.85	-1.70
MR-SAR	66.88	65.67	65.11	64.82	65.60	61.70
Decline	/	-1.21	-1.77	-2.06	-1.28	-5.18

Both features performs not very well on MR-SAR data, mainly because moderate-resolution SAR images cannot provide sufficiently discriminative information. Nevertheless, compared to original NGFs [Fig. 7(b-1)], the propose GTML still achieves the maximum separation of the three subcategories [Fig. 7(b-2)].

3) Confusion Matrix: The confusion matrix of GTML (GTML-A) on SLS task (Table VII) shows that due the subcategories are well separated, the classification *recall* of all subcategories has been greatly improved, especially for the tanker class, increases from 44.20% (Table V) to 70.64% (Table VII) on HR-SAR data, and increases from 55.80% to 76.60% on MR-SAR data.

4) Ablation Study: Table VIII shows the performance decline after a specific optimization component is removed. It shows that the effects of removing MDA, CDA, and MR on the results does not look as significant in the SLS task (Table VIII) as in the ZLS task (Table IV). One possible explanation is that in supervised tasks, the classifier tends to rely on the strong information provided by labels. In contrast, the distribution and structural information provided by the data itself is weakened.

5) Parameter Sensitivity: Fig. 8 shows the parameter sensitivity of GTML (GTML-A in this experiment) in SLS task. The results also demonstrate GTML can achieve optimal classification performance under a wide range of parameter values ({0.001, 0.01, 0.1, 1, 10, 100, 1000}). Comparing Fig. 8(a), (b) with Fig. 6(a), (b), it reveals that in supervised task SLS, the SCL can maintain a relatively stable state while DCL cannot be set too large in terms of DIP. Another reasonable observation is that the parameters are more stable for AIS→MR-SAR than its counterpart in ZLS task as show in Fig. 6, because there are labeled data available in the SLS task.

6) Time Complexity: The time complexity of all algorithms is empirical evaluated by running them on the AIS→HR-SAR/MR-SAR task. Comparing some well-performing methods whose MP exceed 70.00% as shown in Table VI, it reveals that, in contrast, DML-based methods like MMC, DML_eig, SML and ITML have lower time complexity than TL and TML-based methods. The proposed GTML achieves a running time complexity comparable to TL-based method HFA, but far better than other well-performing TML-based methods L-DML and M-DML.

V. CONCLUSION

To cope with various challenges in SAR ship classification problem, we proposed a novel method of GTML, which integrates DIP, DA, and GSP into a unified optimization function, aiming to make full use of their complementarity to improve SAR ship classification performance. In practice, we proposed two simple but effective optimization strategies, termed as GTML-A and GTML-R, to construct optimization function. We also proposed two solutions for two typical real-world application scenarios, that is, ZLS task and SLS task. The experiments conducted on both tasks show that the proposed GTML outperforms most of state-of-the-art methods. Further analysis explained the rationality of the experimental results theoretically, and pointed out that DA and GSP play an important role in the ZLS task, while TD data plays a decisive role in the SLS task. In the future work, we will focus on three aspects:

1) Improving the Ability of Distance Metrics to Process Nonlinear Data: In this study, we adopted the CVX convex optimization toolbox [75] to solve the semidefinite programming problem of objective function. As we know, the time complexity of CVX increases dramatically with the increased dimension of optimization variable (distance metric $\boldsymbol{M}$ in this

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TABLE IX
ABBREVIATIONS OF EXISTING APPROACHES

Abbreviation	Full Name
ACA	adapting component analysis
ARTL	adaptation regularization framework-based transfer learning
A-SVM	adaptive support vector machine
BDA	balanced distribution adaptation
CDLS	cross-domain landmarks selection
CDML	cross-domain metric learning
CORAL	correlation alignment
DAMA	domain adaptation manifold alignment
DAML	domain adaptation metric learning
D-ARTL	discriminative adaptation regularization framework-based transfer learning
DCA	discriminative component analysis
DLC-DA	discriminative label consistent domain adaptation
DML-dc	distance metric learning using difference of convex functions programming
DML-ds	distribution shift metric learning
DML-eig	distance metric learning eigenvalue optimization framework
DMLMJ	distance metric learning through maximization of the Jeffrey divergence
D-SVM	doublet support vector machine
DTDML	decomposition-based transfer distance metric learning
GFK	geodesic flow kernel
GRSC	graph regularized sparse coding
GTML	geometric transfer metric learning
HFA	heterogeneous feature augmentation
ITL	information-theoretical learning
ITML	information-theoretic metric learning
JDA	joint distribution adaptation
JGSA	joint geometrical statistical alignment
LDADA	linear-discriminant-analysis-inspired domain adaptation
L-DML	log-determinant regularized distance metric learning
LGA	latent graph alignment
LMNN	large margin nearest neighbor
LODML	linear ordinal distance metric learning
LSDT	latent sparse domain transfer
LRML	Laplacian regularized metric learning
LRDR	low-rank discriminative reconstruction
LRSR	low-rank sparse representation
M-DML	manifold-determinant regularized distance metric learning
MEDA	manifold embedded distribution alignment
MMC	min-max constraint
MMDT	max-margin domain transforms
MTDML	manifold regularized transfer distance metric learning
MTLF	metric transfer learning framework
NCA	neighbourhood component analysis
PMT-SVM	projective model transfer support vector machine
RDML	regularized distance metric learning
RTML	robust transfer metric learning
SA	subspace alignment
SDA	subspace distribution alignment
SML	sparse metric learning
STL	stratified transfer learning
Sub-SML	similarity metric learning
TCA	transfer component analysis
TJM	transfer joint matching
TKL	transfer kernel learning
TLR	transfer latent representation
TrDCA	trace discriminative component analysis
TSC	transfer sparse coding
T-SVM	triplet support vector machine
VDA	visual domain adaptation
W-JDA	weight joint distribution adaptation

TABLE X
ABBREVIATIONS OF SOME PROPER NOUNS AND ALGORITHMS

Abbreviation	Full name
AIS	automatic identification system
A-MML	accumulative max-min loss
CDA	conditional distribution alignment
DA	domain adaptation
DC	dissimilar constraints
DCL	dissimilar constraints loss
DIP	discriminative information preservation
DML	distance metric learning
DS	domain shift
EL	exponential loss
FN	Frobenius norm
GFK	geodesic flow kernel
GSP	geometric structure preservation
HL	hinge loss
HR-SAR	high-resolution SAR
HSIC	Hilbert-Schmidt independence criterion
IR	instance re-weighting
IRF	inter-class repulsive force
ISM	inter-class scatter matrix
JD	Jeffrey divergence
JDA	joint distribution adaptation
k-NN	k-nearest neighbor
L_2	L_2 norm
LC	low-rank constraints
LCR	label consistent regression
LD	Log-determinant divergence
LE	linear entropy
LMP	large-margin principle
LOE	leave-one-out error
LTC	local triple constraints
MC	max-margin classifier
MDA	marginal distribution alignment
MI	mutual information
MKL	multi-kernel learning
MMD	maximum mean discrepancy
MP	mean performance
MML	max-min loss
MR	manifold regularization
MR-SAR	moderate-resolution SAR
NGFs	naive geometric features
PC	pairwise constraints
PSD	positive semi-definite
RBF	radial basis function
RC	reconstruction constraints
RL	ramp loss
R-MML	ratio max-min loss
SAR	synthetic aperture radar
Sc	sparse constraints
SC	similar constraints
SCL	similar constraints loss
SD	source domain
SDA	supervised domain adaptation
SDIP	source discriminative information preservation
SDM	subspace divergence minimization
SDP	semi-definite programs
SLS	scarce labeled samples
SRM	structural risk minimization
STML	supervised transfer metric learning
SVM	support vector machine
TC	triple constraints
TD	target domain
TL	transfer learning
TML	transfer metric learning
TVM	target variance maximization
UDA	unsupervised domain adaption
UTML	unsupervised transfer metric learning
WSM	within-class scatter matrix
ZLS	zero labeled sample

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study). Therefore, it limits the application of more discriminative high-dimensional feature representation, and it also limits the application of kernel techniques to solve nonlinear problems. In the next work, we hope to develop more effective optimization method or extend GTML to deep TML methods.

2) Expanding GTML to More SDs: In this study, we applied AIS as the SD to assist the ship classification in SAR domain. AIS as the SD has two obvious deficiencies. One is significant DS to the SAR domain which brings great difficulties for various TL methods. The other deficiency is that limited by the information that AIS can provide, the transferable features between AIS domain and SAR domain are restricted to NGFs. As a result, more discriminative SAR image features cannot be utilized, which prevents further improvement in ship classification performance. In the next work, we will further explore the feasibility of the GTML in other more suitable SD (such as the optical image domain), or even using multiple SDs in combination.

3) Combining Feature Transfer and Parameter Transfer: In addition, the ground-breaking researches of TL for ship classification in SAR images [16], adaptive support vector machine (A-SVM) (76.03%) [76] and projective model transfer support vector machine (PMT-SVM) (75.82%) [77], obtain well MP in the SLS task, which proved the importance of transferring prior parameter knowledge. In the next work, we hope to design more complete TL methods, which combine feature transfer and parameter transfer, for example, combining distribution alignment or manifold consistency alignment with A-SVM and PMT-SVM.

APPENDIX

For the convenience of reading, we list the names of existing approaches, some proper nouns and algorithms mentioned in this article, and their corresponding abbreviations in the Table IX and Table X. The abbreviations are arranged in the first alphabetical order.

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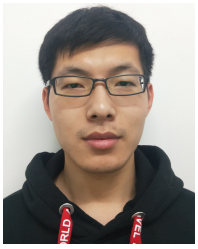
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Yongjie Xu (Student Member, IEEE) received the B.S. and M.S. degrees from the Beijing University of Chemical Technology, Beijing, China, in 2017 and 2020, respectively. He is currently pursuing the Ph.D. degree in electronic science and technology with the cooperative training of Zhejiang University, Hangzhou, China, and Westlake University, Hangzhou.

His research interests include SAR/optical remote sensing image classification and machine learning.



Haitao Lang (Member, IEEE) received the B.S. and M.S. degrees in applied physics and optical engineering from the Ocean University of China, Qingdao, China, in 2000 and 2003, respectively, and the Ph.D. degree in optical engineering from the Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Shanghai, China, in 2006.

He is currently a Professor with the College of Mathematics and Physics, Beijing University of Chemical Technology, Beijing, China. His research interests focus on developing novel machine learning/deep learning algorithms for object detection, recognition, and classification in optical/SAR remote sensing imagery/data.

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